

AWS Cloud Stack for the AWS Learning Portal

Project: AWS Learning Portal

Course: CS 79C Compute Engines in Amazon Web Services

Instructor: Vicky Tanya Seno

Student: Siarhei Hancharou

Student ID: 1712666

1. Core AWS Services

Compute & Application Layer:

- Amazon EC2 - hosting frontend and backend

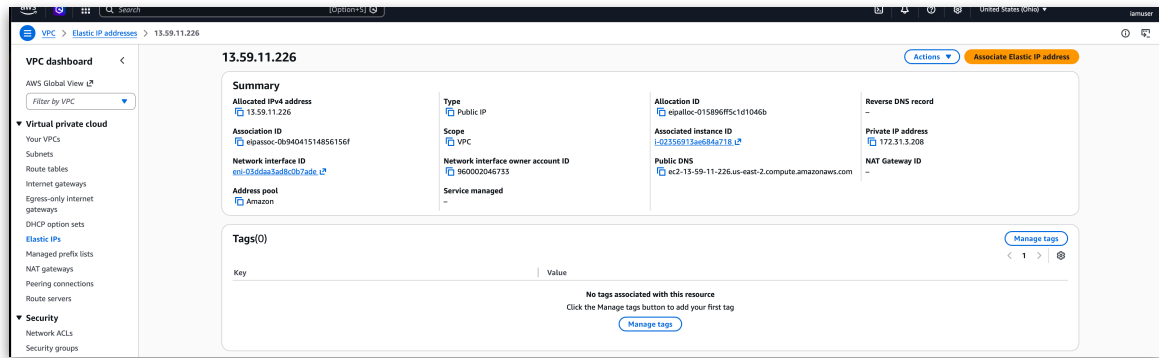
The screenshot displays the AWS Management Console for an EC2 instance with ID `i-02356913ae684a718`. The instance is running on the `cs79c-ec2-role` IAM role, using the `al2023-ami-2023.10.20260302.1-kernel-6.1-x86_64` AMI. It is located in the `us-east-2` region, specifically in the `us-east-2a` Availability Zone. The instance has a public IP address of `13.59.11.226` and is part of the `ec2-13-59-11-226-us-east-2-compute` Elastic IP pool. The instance is running on the `t2.micro` instance type, which is associated with the `vpc-0ed8d190686daed34` VPC and the `subnet-0f5706de48f739a82` subnet. The instance is currently in the `Running` state. The console also shows the instance's lifecycle, including its launch time, and the state transition message: `Client.UserInitiatedShutdown: User initiated shutdown`.

```
[ec2-user@ip-172-31-3-208 ~]$ pm2 list
```

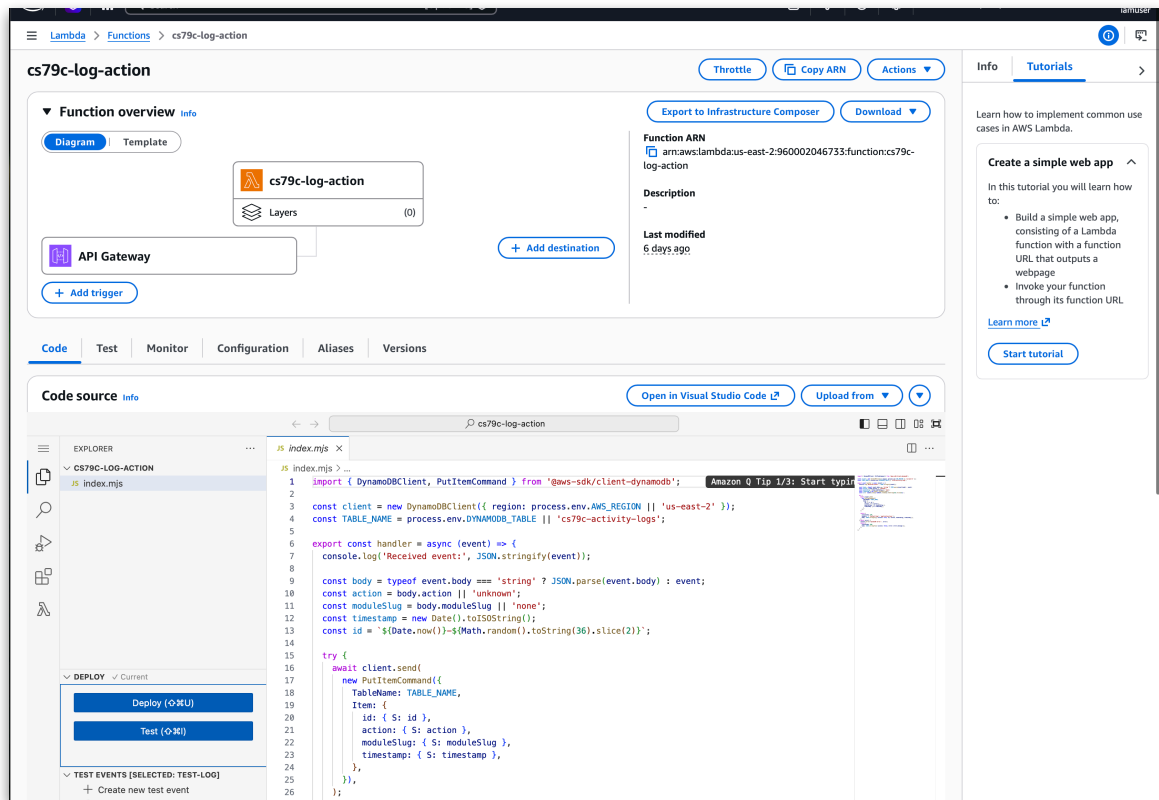
id	name	mode	u	status	cpu	memory
0	cs79c-backend	fork	0	online	0%	74.6mb
1	cs79c-frontend	fork	0	online	0%	54.8mb

```
[ec2-user@ip-172-31-3-208 ~]$
```

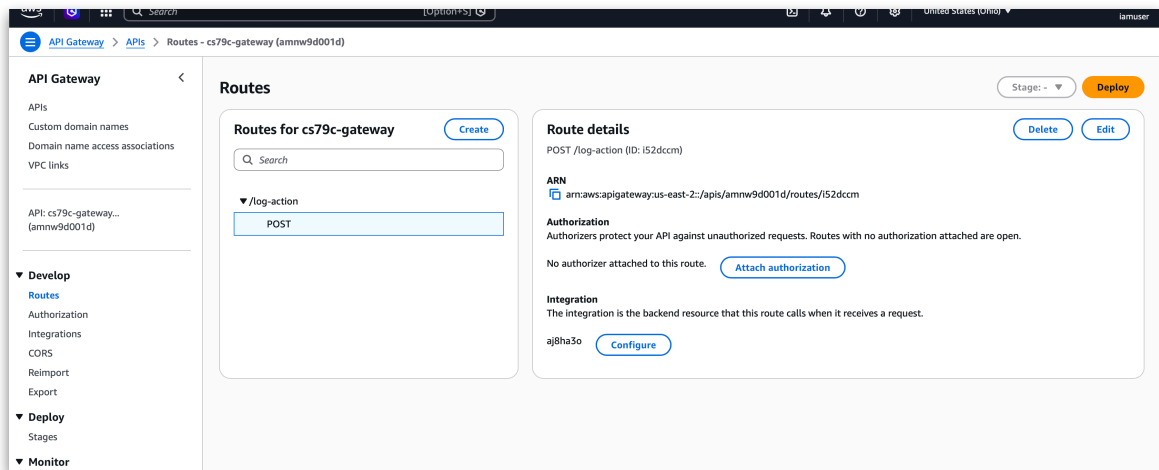
- Configured a static Public IPv4 13.59.11.226 using an Elastic IP for stable public access



- AWS Lambda - serverless processing

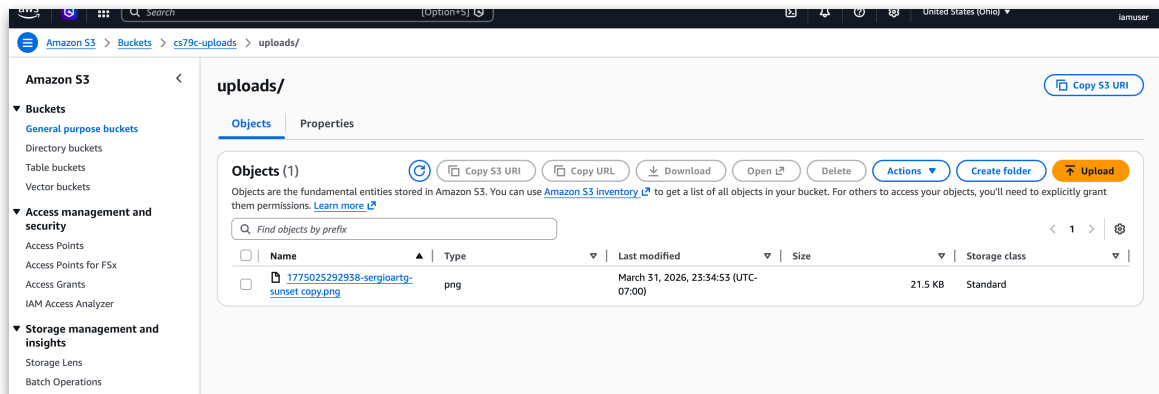


- Amazon API Gateway - API routing

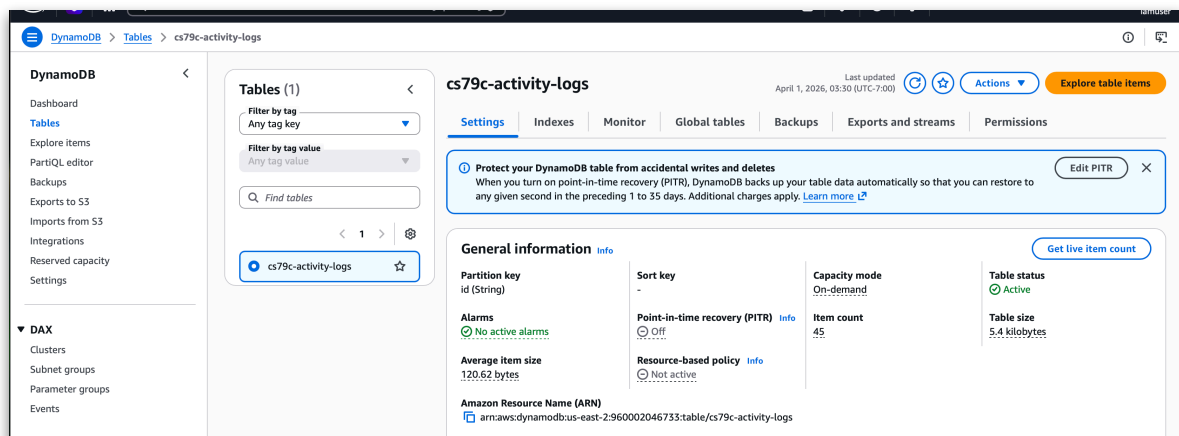


Storage & Database:

- Amazon S3 - file storage



- Amazon DynamoDB - NoSQL database for logs



Networking & Security:

- Amazon VPC (default VPC networking for the current EC2 deployment)
- Public Subnets (Multi-AZ)

The screenshot displays the AWS Management Console interface for a VPC subnet. The left sidebar shows the navigation menu with categories like VPC dashboard, Virtual private cloud, Security, and PrivateLink and Lattice. The main content area is titled 'subnet-0f5706de48f739a82' and contains several sections: Details, Flow logs, and various tabs for configuration. The Details section provides comprehensive information about the subnet, including its ID, ARN, IP address ranges, availability zone, and associated VPC. The Flow logs section shows a search bar and a table for logging network traffic.

Subnet ID	Subnet ARN	State	Block Public Access
subnet-0f5706de48f739a82	arn:aws:ec2:us-east-2:960002046733:subnet/subnet-0f5706de48f739a82	Available	Off

IPV4 CIDR	Available IPV4 addresses	IPV6 CIDR	Route table
172.31.0.0/20	4090	-	rtb-0fee188e15097f52e

Availability Zone	VPC	Auto-assign customer-owned IPV4 address	Auto-assign customer-owned IPV6 address
us-east-2a	vpc-0ed8d190686d4ed34	No	No

Default subnet	Customer-owned IPV4 pool	IPV6-only	DNS64
Yes	-	No	Disabled

Outpost ID	Hostname type	Owner
-	IP name	960002046733

Flow logs
No flow logs found

- Security Groups

The screenshot shows the AWS Management Console for a security group. The left sidebar lists navigation options under EC2, including Dashboard, Events, Instances, Images, and Elastic Block Store. The main content area is titled 'sg-007dc805683d3894d - cs79c-backend-sg' and includes a Details section, Inbound rules, and various tabs for configuration. The Details section provides information about the security group's name, ID, description, and VPC. The Inbound rules section displays a table of rules that allow traffic to the instances.

Security group name	Security group ID	Description	VPC ID
cs79c-backend-sg	sg-007dc805683d3894d	CS79C backend - allow SSH and API port 3001	vpc-0ed8d190686d4ed34

Inbound rules (4)
☐ - sgr-086780f9c386faf9b IPv4 Custom TCP TCP 3001 0.0.0.0/0 NestJS-API
☐ - sgr-010d43bb0180aaa4e IPv4 Custom TCP TCP 3000 0.0.0.0/0 -
☐ - sgr-0ccbfef986f8a76b01 IPv4 HTTP TCP 80 0.0.0.0/0 HTTP
☐ - sgr-01005ca03e35aa426 IPv4 SSH TCP 22 0.0.0.0/0 SSH

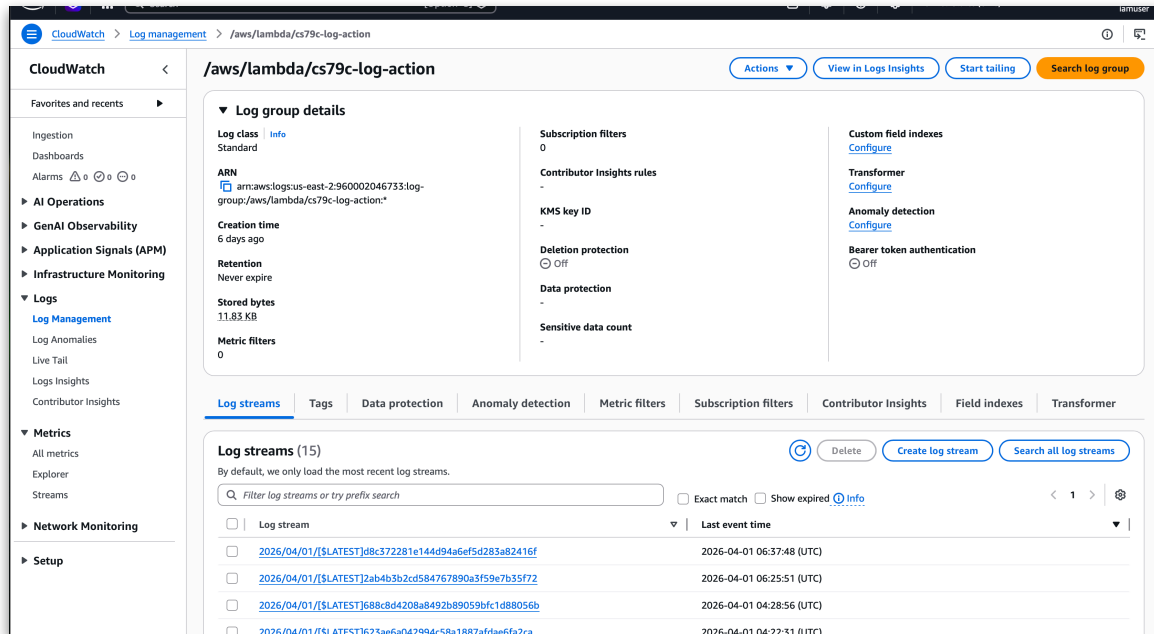
- IAM Roles / Instance Profiles

The screenshot displays the AWS Management Console for an IAM role. The left sidebar shows the navigation menu for Identity and Access Management (IAM), including Access Management, Access reports, and IAM Identity Center. The main content area is titled 'cs79c-ec2-role' and includes a Summary section, Permissions policies, and various tabs for configuration. The Summary section provides information about the role's creation date, last activity, and instance profile. The Permissions policies section shows a table of policies attached to the role.

Policy name	Type	Attached entities
AmazonDynamoDBFullAccess	AWS managed	1
AmazonS3FullAccess	AWS managed	1
AWSLambdaRole	AWS managed	1

2. Monitoring & DevOps

- Amazon CloudWatch - monitoring and logging



- AWS SDK for JavaScript v3 - integration layer

```
[ec2-user@ip-172-31-3-208 ~]$ cd /home/ec2-user/app
[ec2-user@ip-172-31-3-208 app]$ grep "@aws-sdk" backend/package.json
  "@aws-sdk/client-dynamodb": "^3.1017.0",
  "@aws-sdk/client-lambda": "^3.1017.0",
  "@aws-sdk/client-s3": "^3.1017.0",
  "@aws-sdk/s3-request-presigner": "^3.1017.0",
[ec2-user@ip-172-31-3-208 app]$
```

3. Operating System

- Amazon Linux 2023

4. Architecture Summary

Client → EC2 (Frontend)
Frontend → Backend (EC2)
Backend → S3
Backend → API Gateway → Lambda
Lambda → DynamoDB
CloudWatch → Monitoring